DILTIAZEM

Avoid in HFrEF

Bradycardia

CHEMICAL NAME (2S,3S)-5-[2-(dimethylamino)ethyl]-2-(4-methoxyphenyl)-4-oxo-2,3,4,5-tetrahydro-1,5-benzothiazepin-3-yl acetate (hydrochloride)

CLASS Non-dihydropyridine calcium-channel blocker (benzothiazepine)

TRADE NAMES Cardizem · Tildiem · Herbesser · Altiazem

THERAPEUTIC USES / INDICATIONS

- Rate control in atrial fibrillation/flutter (IV or oral).
- Angina (stable, vasospastic).
- Hypertension.
- Anal fissure (topical 2% cream).

ADMINISTRATION

- **PO:** MR tablets/capsules swallowed whole; different brands not interchangeable.
- **IV:** 0.25 mg/kg over 2 min with ECG monitoring; infusion for maintenance.
- Topical ointment to anal margin twice daily.

DOSAGES

- **Angina/hypertension:** 60 mg three times daily or MR 90–360 mg/day.
- **AF rate (IV):** 0.25 mg/kg (~20 mg) over 2 min; repeat 0.35 mg/kg after 15 min; infusion 5–15 mg/h.
- **AF rate (PO):** 120–360 mg/day.
- Elderly: start 60 mg twice daily.

PHARMACOKINETICS

- **Onset:** PO 30–60 min; IV 3 min. **Duration:** 6–8 h (IR); 24 h (MR).
- Metabolised by CYP3A4 (and inhibits it).
- Excreted in urine and bile. **t½:** 3–5 h.

PHARMACODYNAMICS

- Blocks **L-type calcium channels** in heart and vessels.
- Slows SA node and AV node conduction → ↓ HR; ↓ contractility (moderate).
- Coronary and peripheral vasodilation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bradycardia, dizziness.
- Headache, flushing, ankle oedema.
- Constipation (less than verapamil).

ADVERSE EFFECTS (SERIOUS)

- AV block, sinus arrest.
- Hypotension.
- Worsening heart failure.
- Rash, SJS (rare); hepatitis.

CONTRAINDICATIONS & CAUTIONS

- Sick sinus syndrome, 2nd/3rd-degree AV block (without pacemaker).
- Heart failure with reduced EF; cardiogenic shock.
- WPW with AF.
- Severe hypotension; IV with IV β-blocker.

KEY INTERACTIONS

- **β-blockers:** heart block, bradycardia, HF.
- **Simvastatin, atorvastatin, ciclosporin, tacrolimus, carbamazepine:** ↑ levels (CYP3A4 inhibition).
- **Digoxin:** ↑ levels.
- **Clopidogrel/DOACs:** altered levels.

NURSING MANAGEMENT

- Check apical HR and BP before each dose; hold if HR <50–60.
- IV: continuous ECG and BP monitoring.
- Monitor for signs of heart failure.
- Review interacting drugs.
- Same brand for MR formulations.

PATIENT EDUCATION

- Swallow MR tablets whole; stick to the same brand.
- Report slow pulse, fainting, breathlessness, swelling.
- Avoid grapefruit juice.

KEY POINTS

- Non-DHP CCB; slows AV node.
- Avoid in HFrEF and with IV β-blockers.
- AF rate control; angina.
- CYP3A4 inhibitor.

SOURCES CHECKED **[319]** ADIZEM-SR capsules 120 mg (UK SmPC) **[320]** Diltiazem Hydrochloride, IV (US label) **[321]** Diltiazem Hydrochloride, oral (US label)

Full references p. 432–457